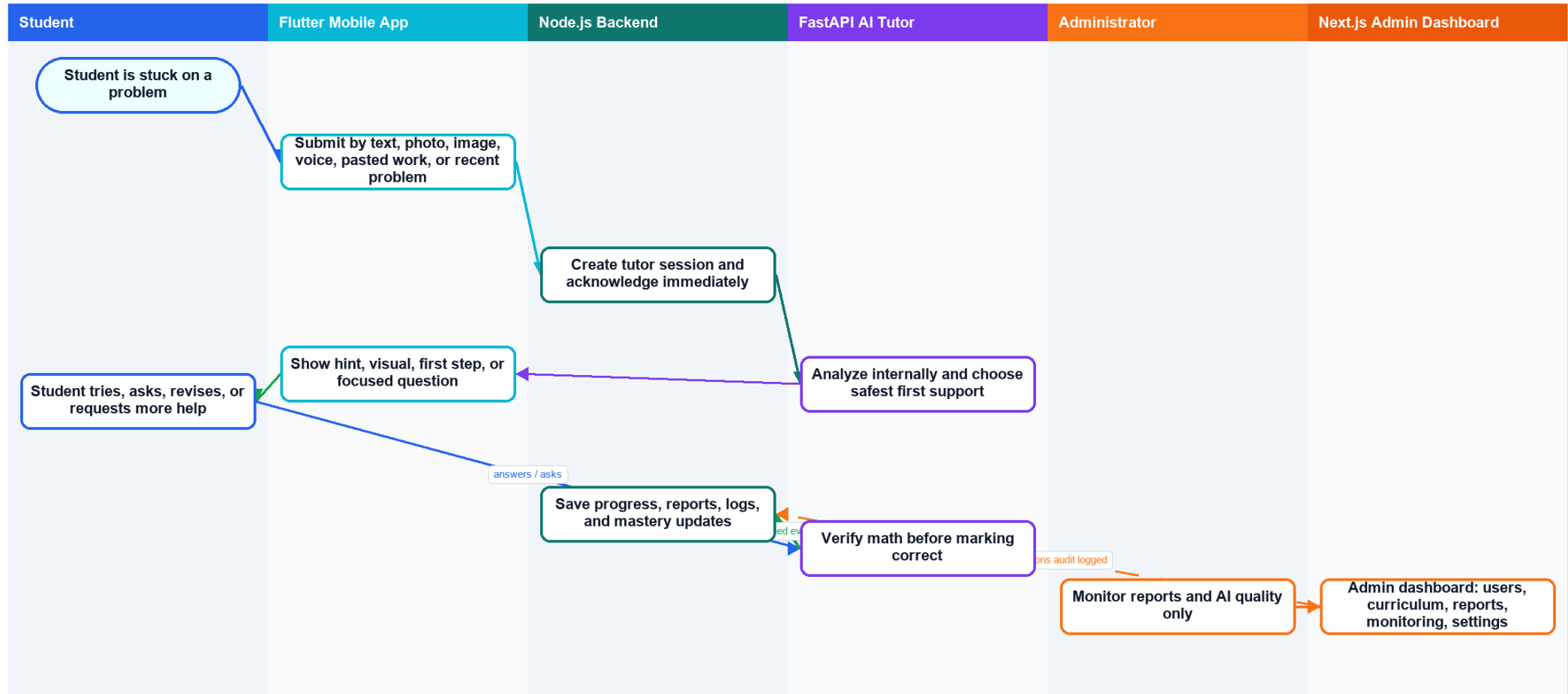


1. System Overview

Student-first journey: stuck problem to useful help, verified answer, practice, and progress.

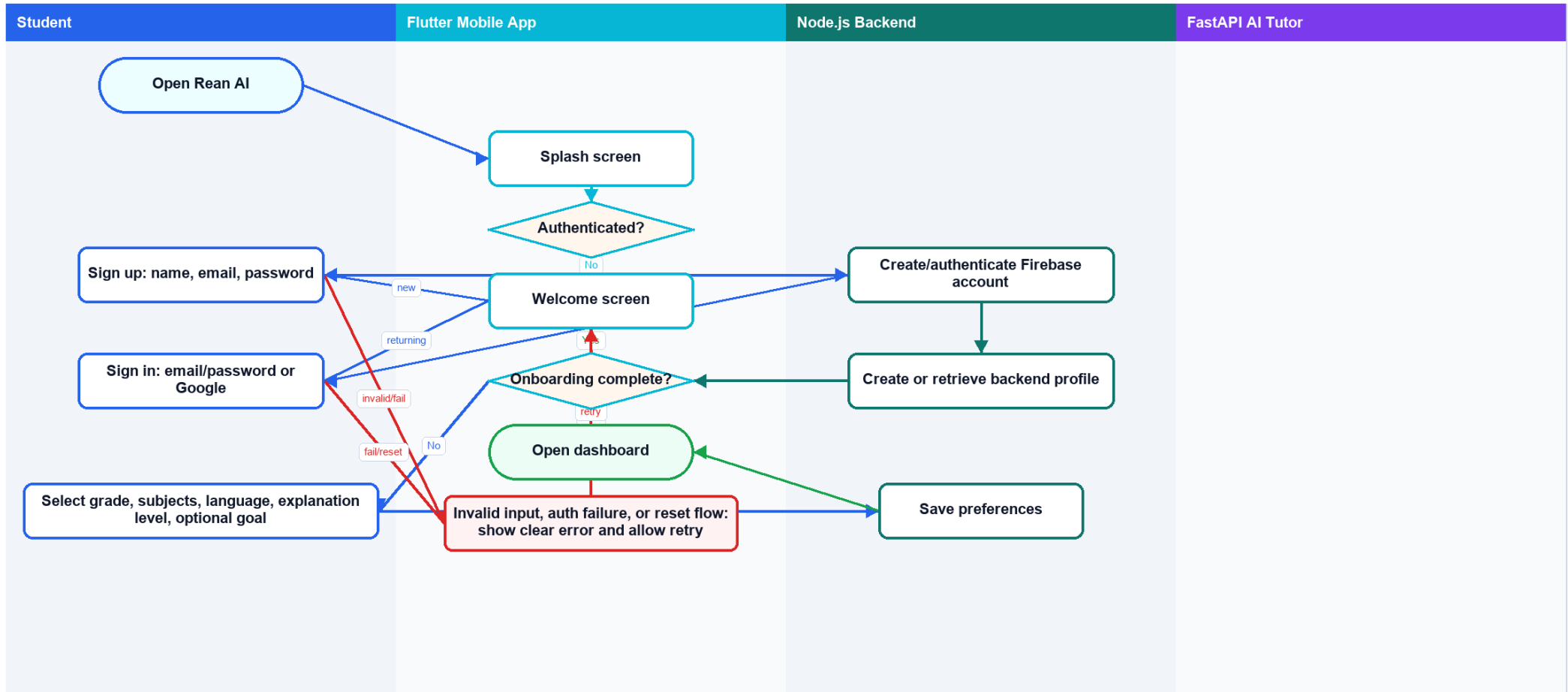


Implementation notes

- No long visible setup sits between problem submission and first useful help.
- The full solution is available, but not the default first response.
- The admin flow is intentionally small for MVP.

2. Authentication And Onboarding

Fast account entry, short setup, then dashboard. No unnecessary personal data before learning.

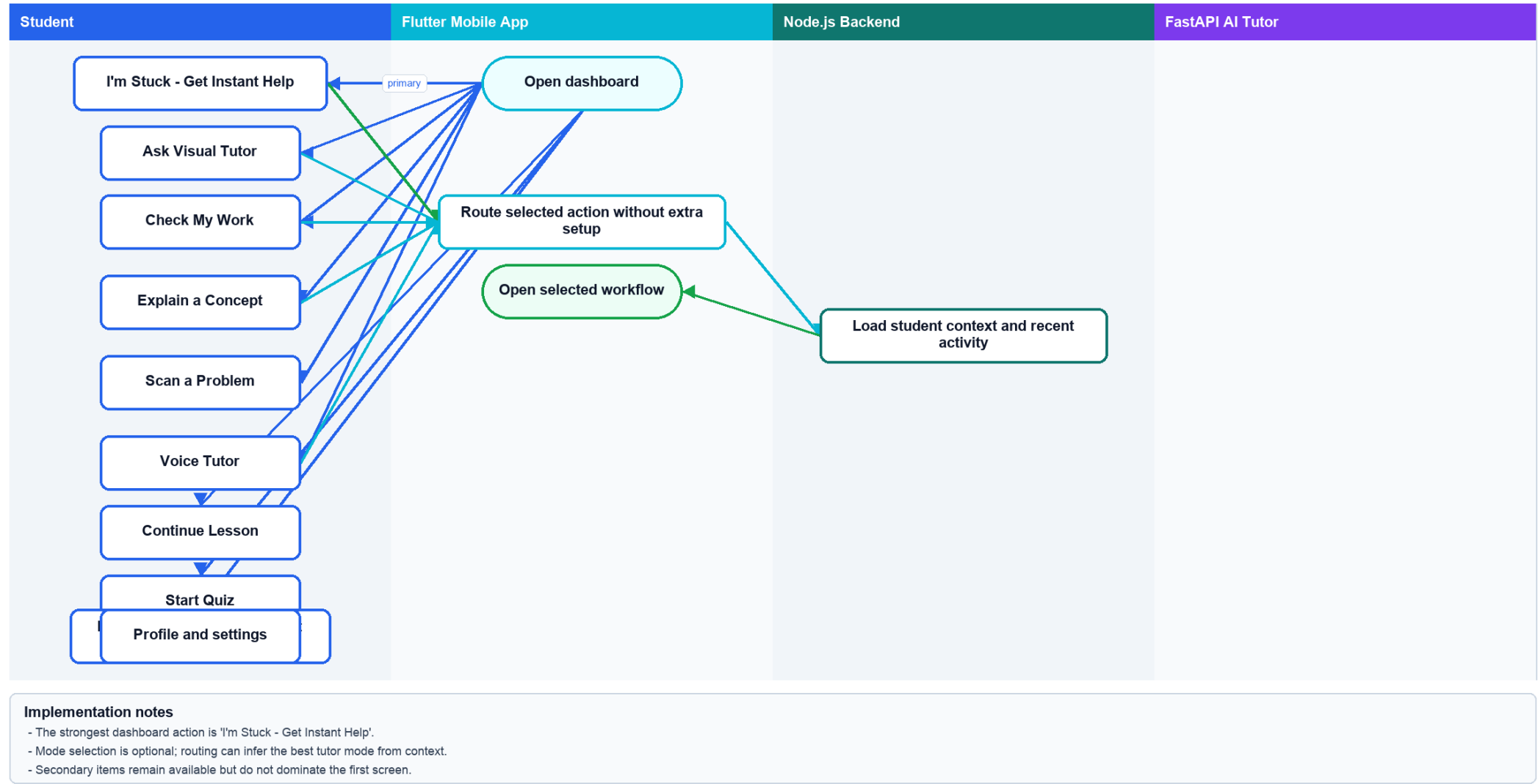


Implementation notes

- Forgot password returns to sign in after reset-link confirmation.
- Google sign-in creates or retrieves the backend profile after Firebase authentication.
- Onboarding remains limited to learning-critical preferences.

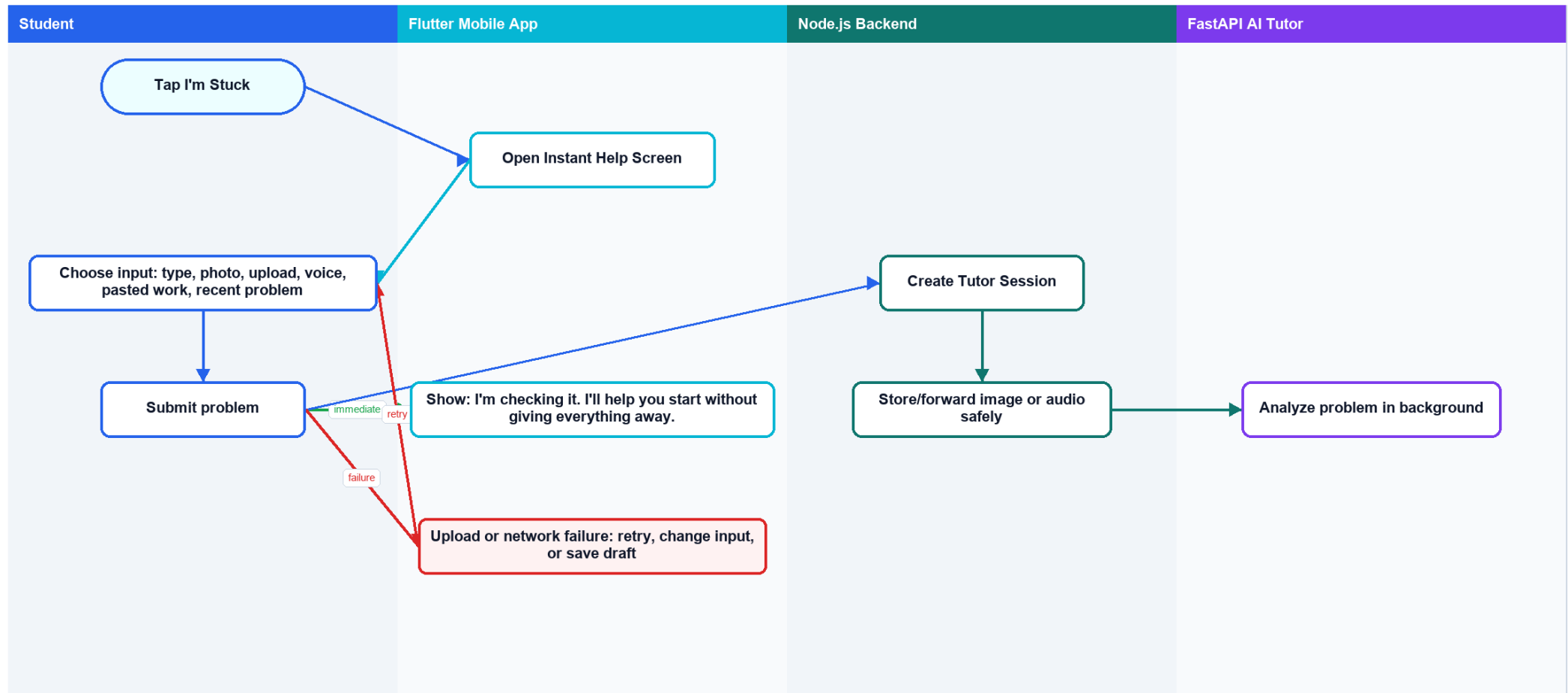
3. Student Dashboard Navigation

Dashboard prioritizes instant learning actions over passive content browsing.



4. Instant Help Entry Flow

Submit quickly, acknowledge immediately, analyze in background.

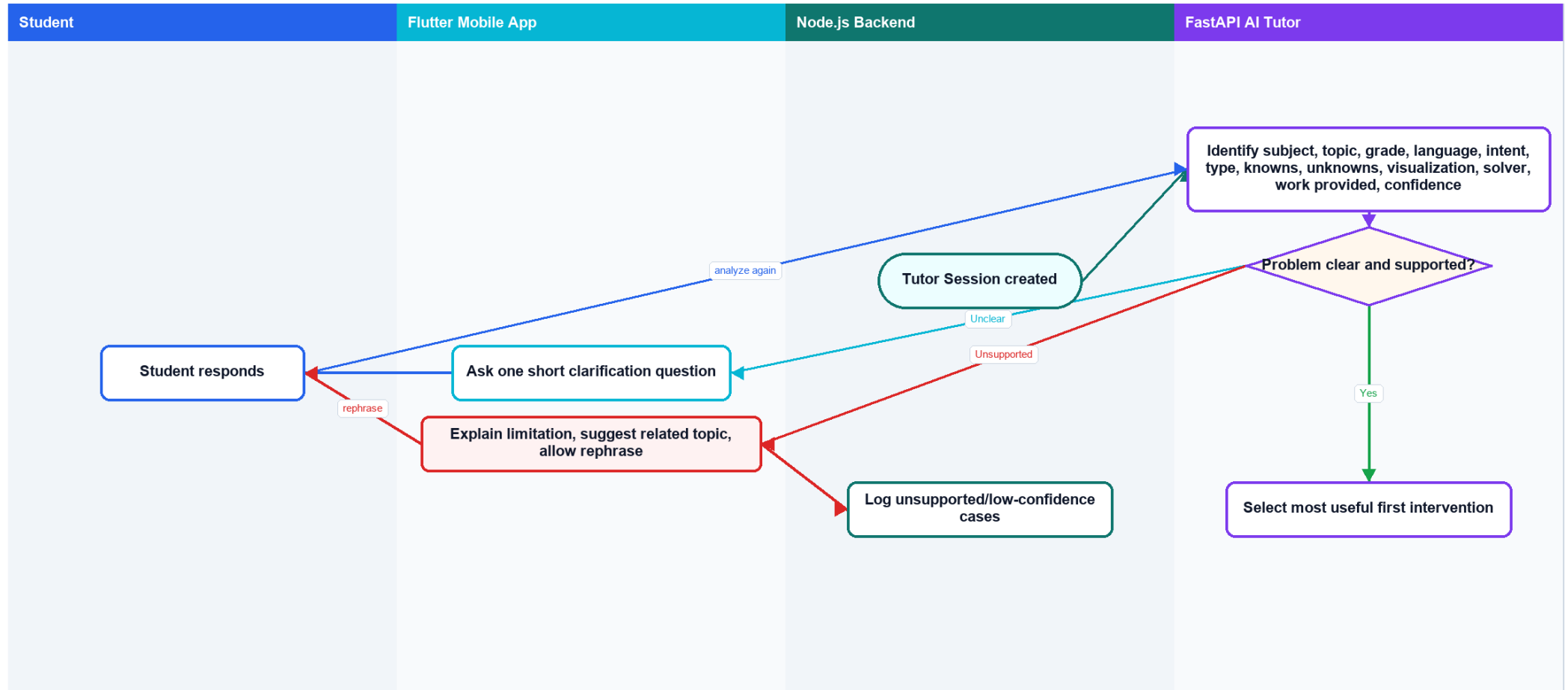


Implementation notes

- Visible acknowledgement appears before the internal AI pipeline finishes.
- Input can include attempted work, enabling Check My Work automatically.
- No raw audio is kept permanently unless retention policy explicitly allows it.

5. Background Question Analysis

Internal classification supports the first useful intervention but does not become a student-facing setup wizard.

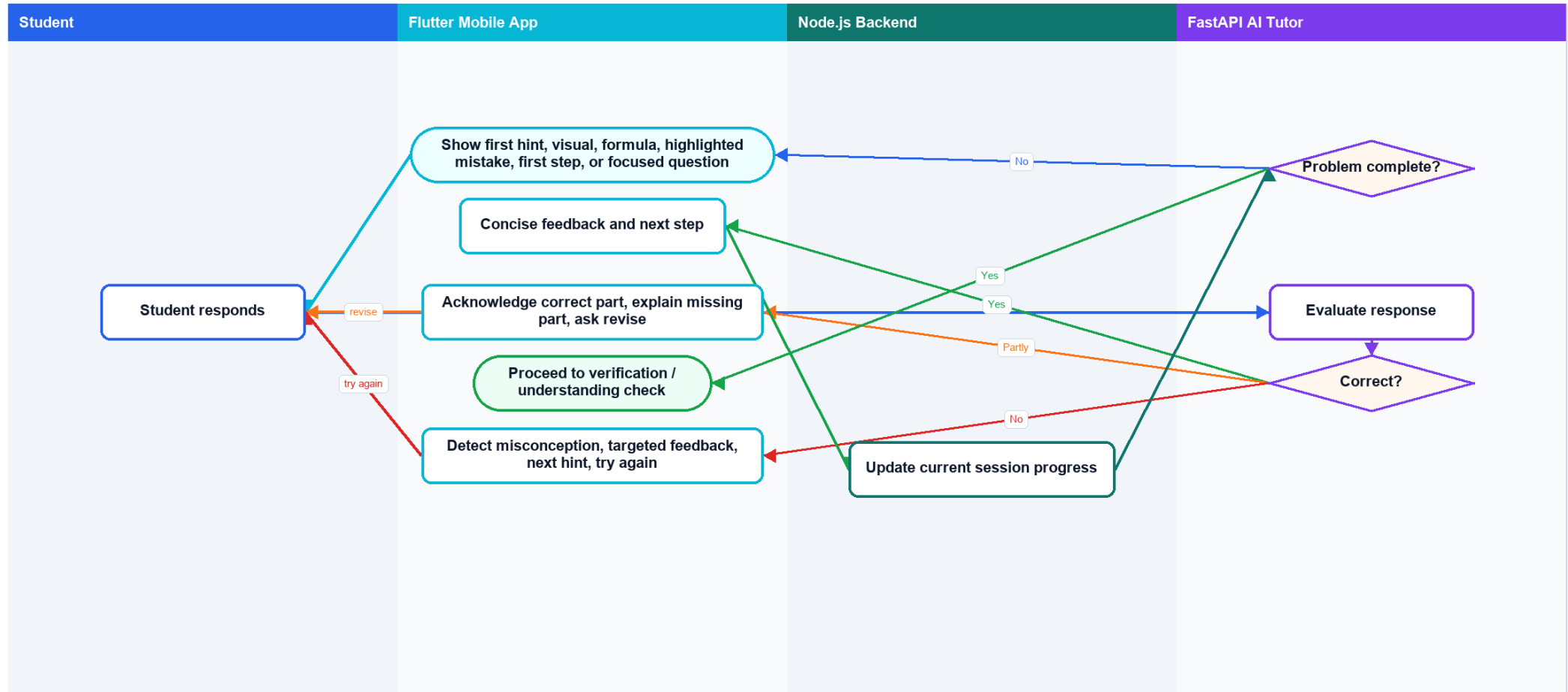


Implementation notes

- Clarification is one question at a time.
- Unsupported problems must not receive fabricated answers.
- Confidence thresholds decide clarify, safe limitation, or tutor response.

6. Interactive Visual Tutor Loop

One short message, one matching visual or formula, one focused question, then adaptive evaluation.

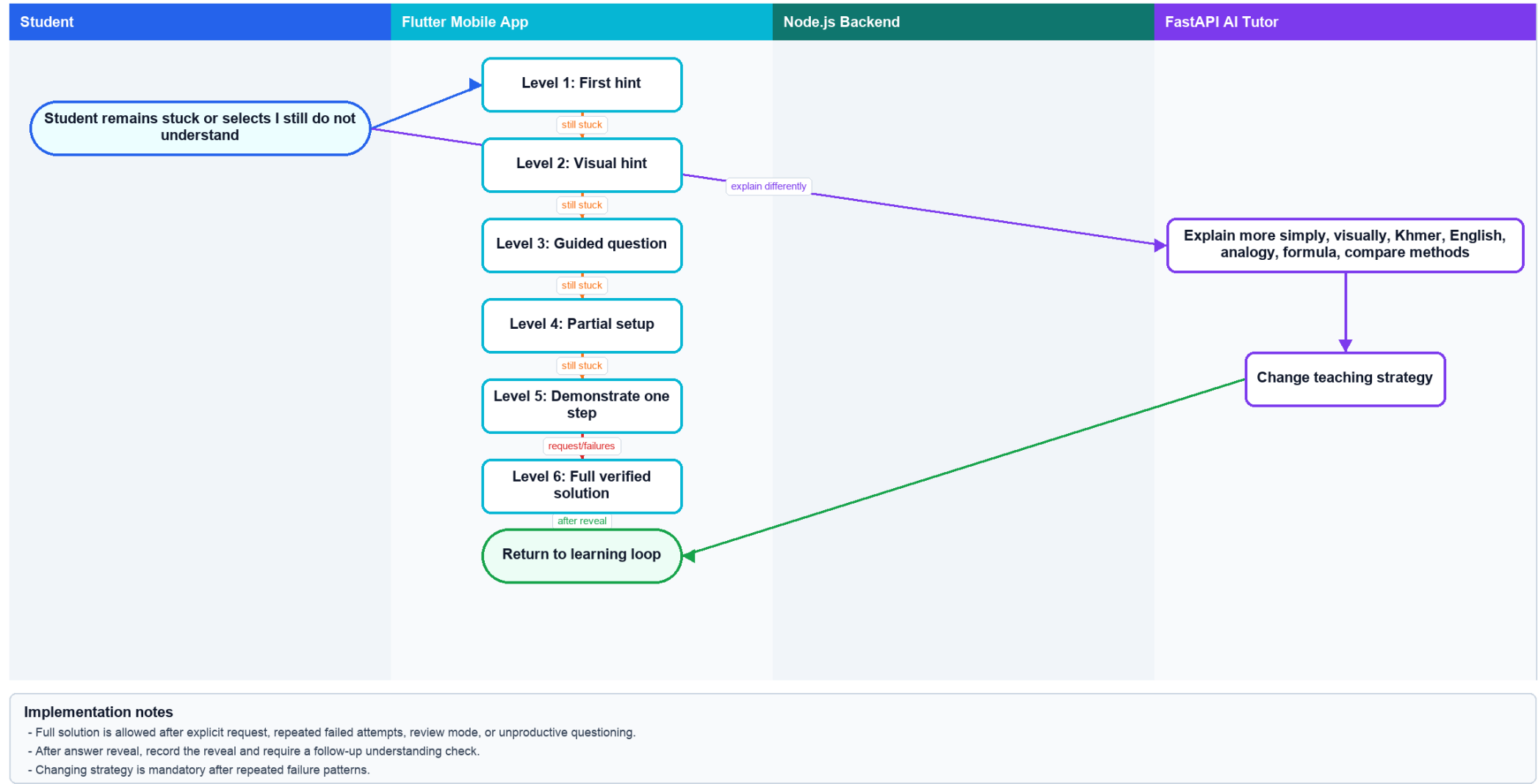


Implementation notes

- The AI must not repeat the same explanation word-for-word.
- Student response quality controls the next tutor step.
- Visuals are attached when they improve understanding, not as decoration.

7. Progressive Hint And Explain Differently

Help becomes more direct when the student remains stuck, while alternative teaching strategies prevent repetition.

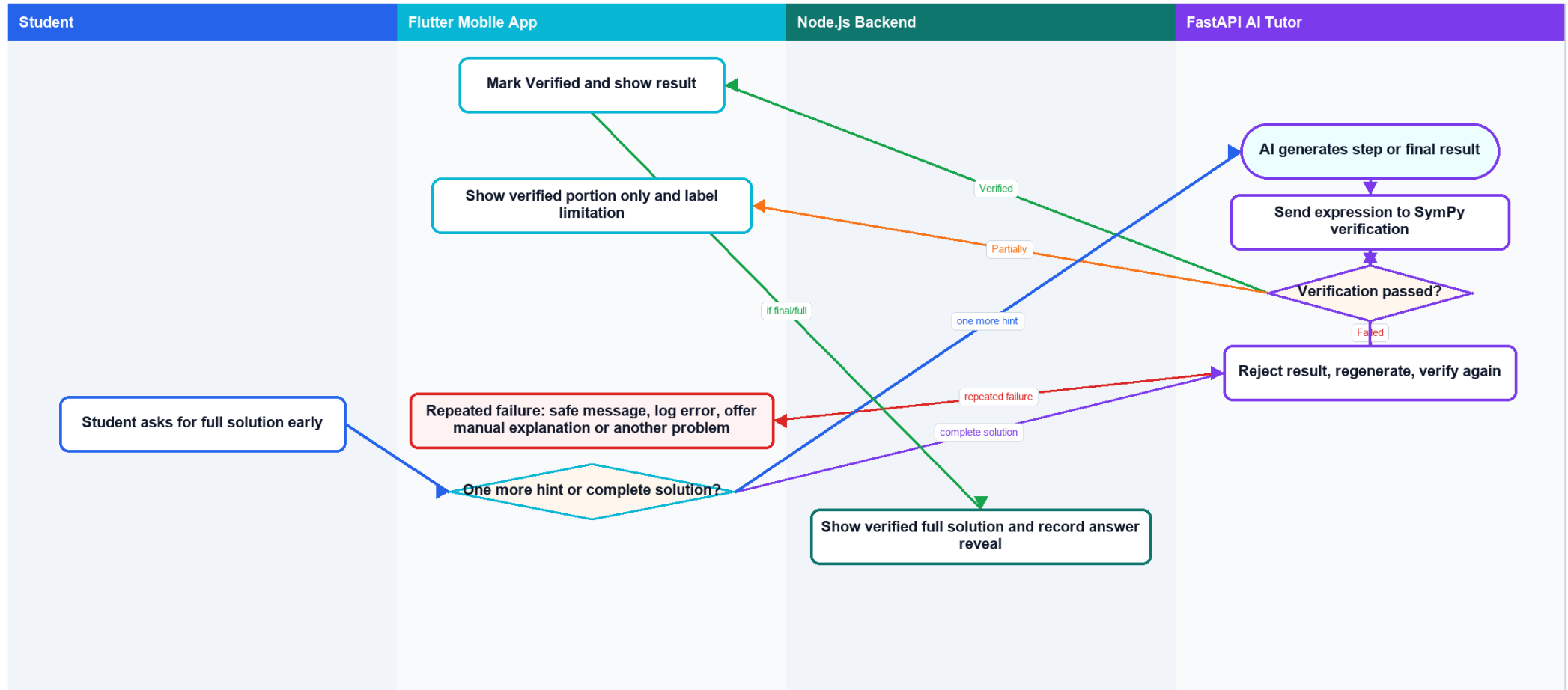


Implementation notes

- Full solution is allowed after explicit request, repeated failed attempts, review mode, or unproductive questioning.
- After answer reveal, record the reveal and require a follow-up understanding check.
- Changing strategy is mandatory after repeated failure patterns.

8. Mathematical Verification And Answer Reveal

No mathematical result is labelled correct until verification passes or limitations are clearly shown.

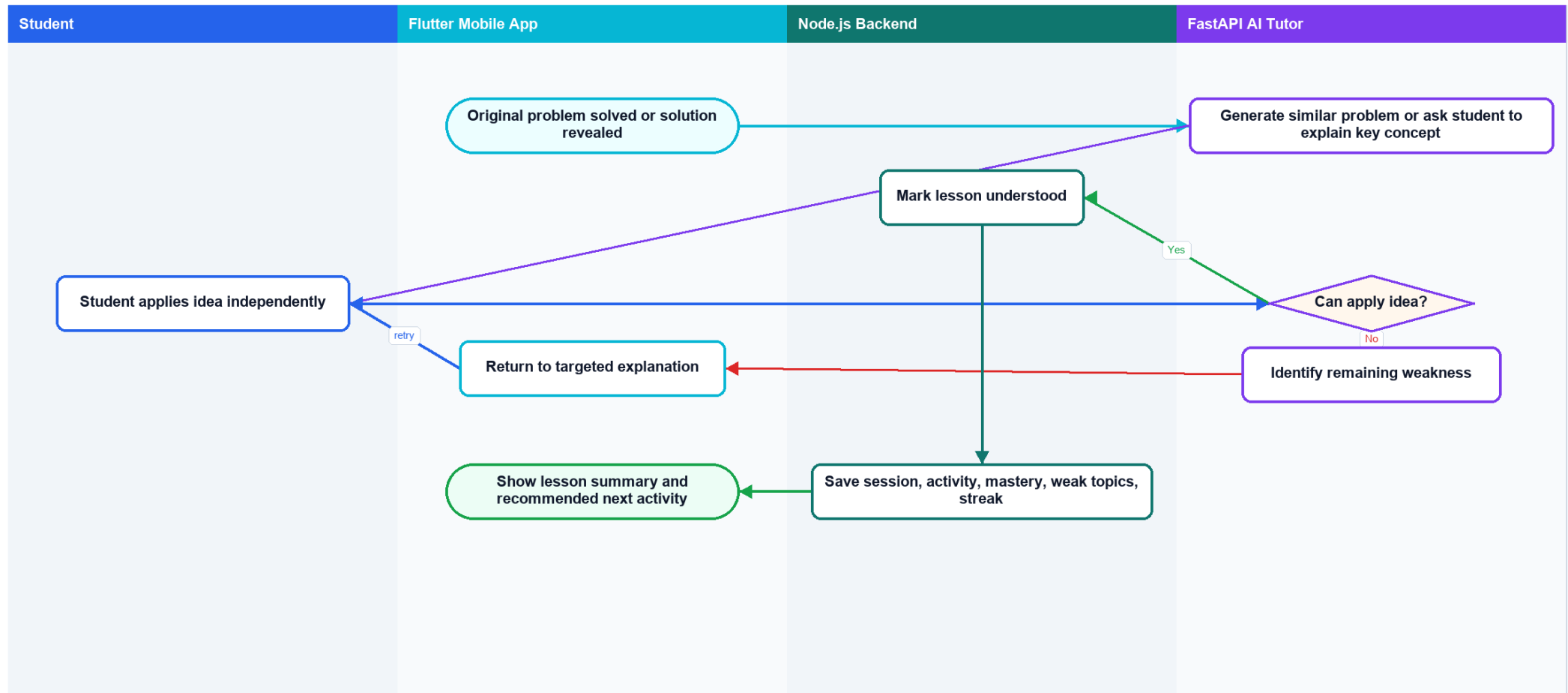


Implementation notes

- Failed verification triggers regeneration or a safe failure path.
- Unverified answers are never presented confidently.
- The product prevents empty copying without blocking answers forever.

9. Understanding Check And Progress Update

Completion requires demonstrated understanding, not just seeing the final answer.

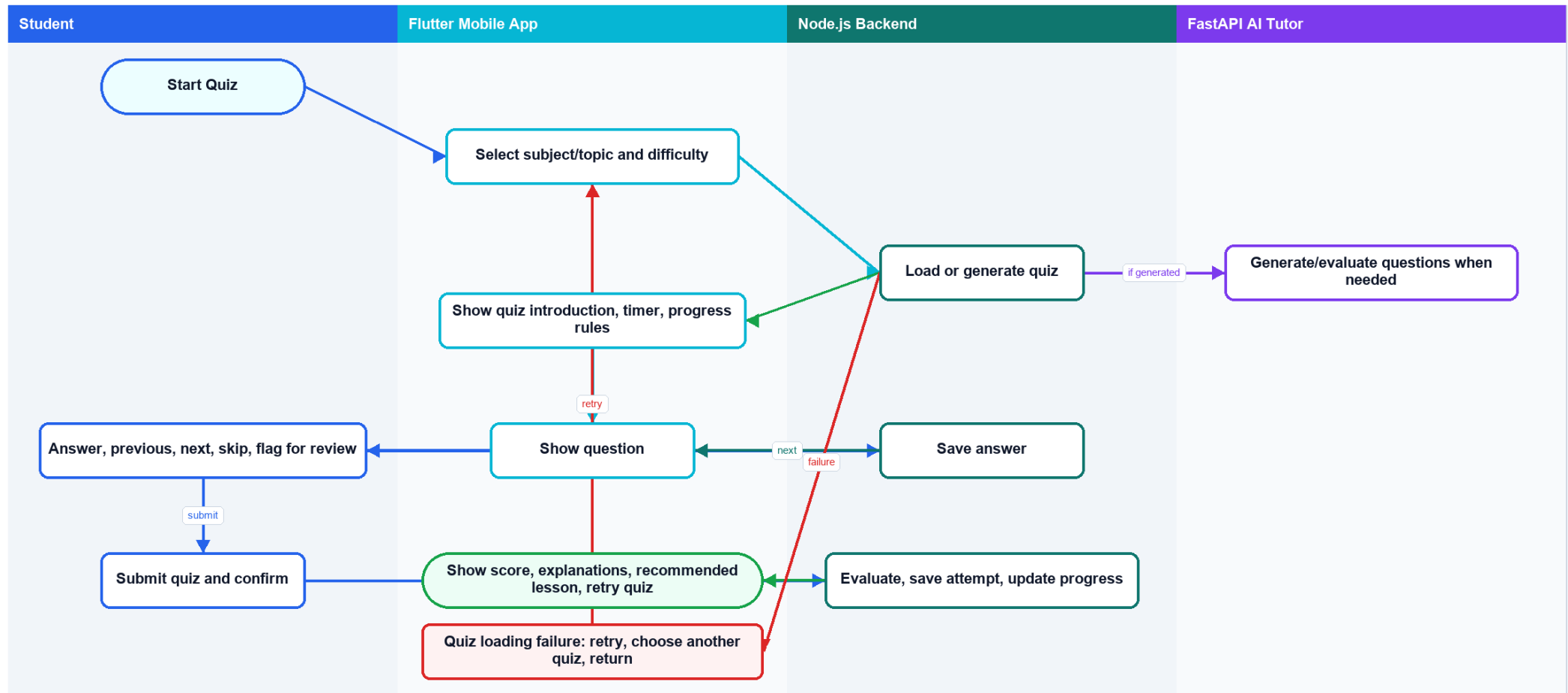


Implementation notes

- Lesson summary includes concept, problem, hints, mistakes, verification status, practice result, and next activity.
- Progress updates after valid learning activity only.
- Save failures show retry and keep a local recoverable session state.

10. Quiz Flow

Quizzes can start from dashboard, completed lesson, weak topic, or recommendation.

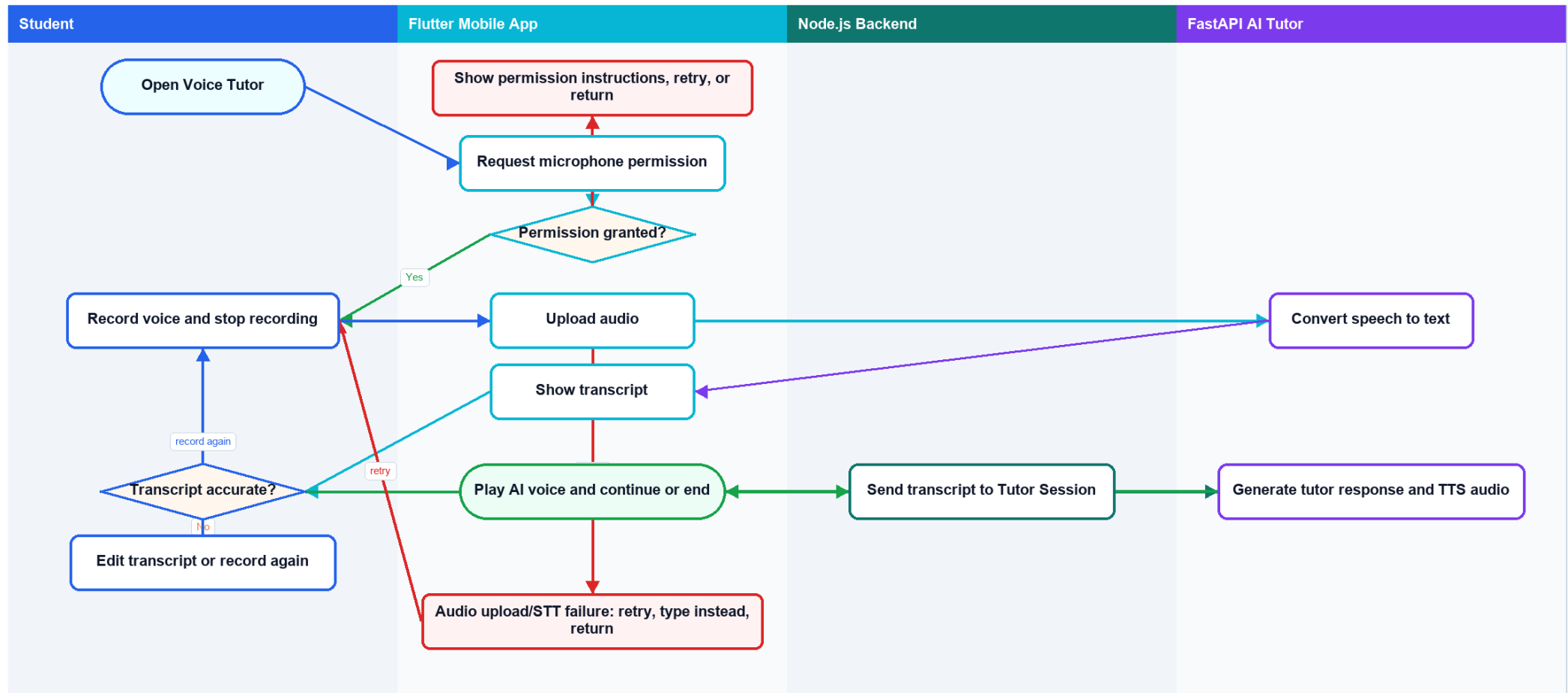


Implementation notes

- Quiz answer evaluation can identify misconceptions and update topic mastery.
- Generated quizzes should inherit the verification rule where math results are involved.
- Students can skip and flag questions without losing flow.

11. Voice Tutor Flow

Voice input follows the same visual tutoring and hint rules after transcript confirmation.

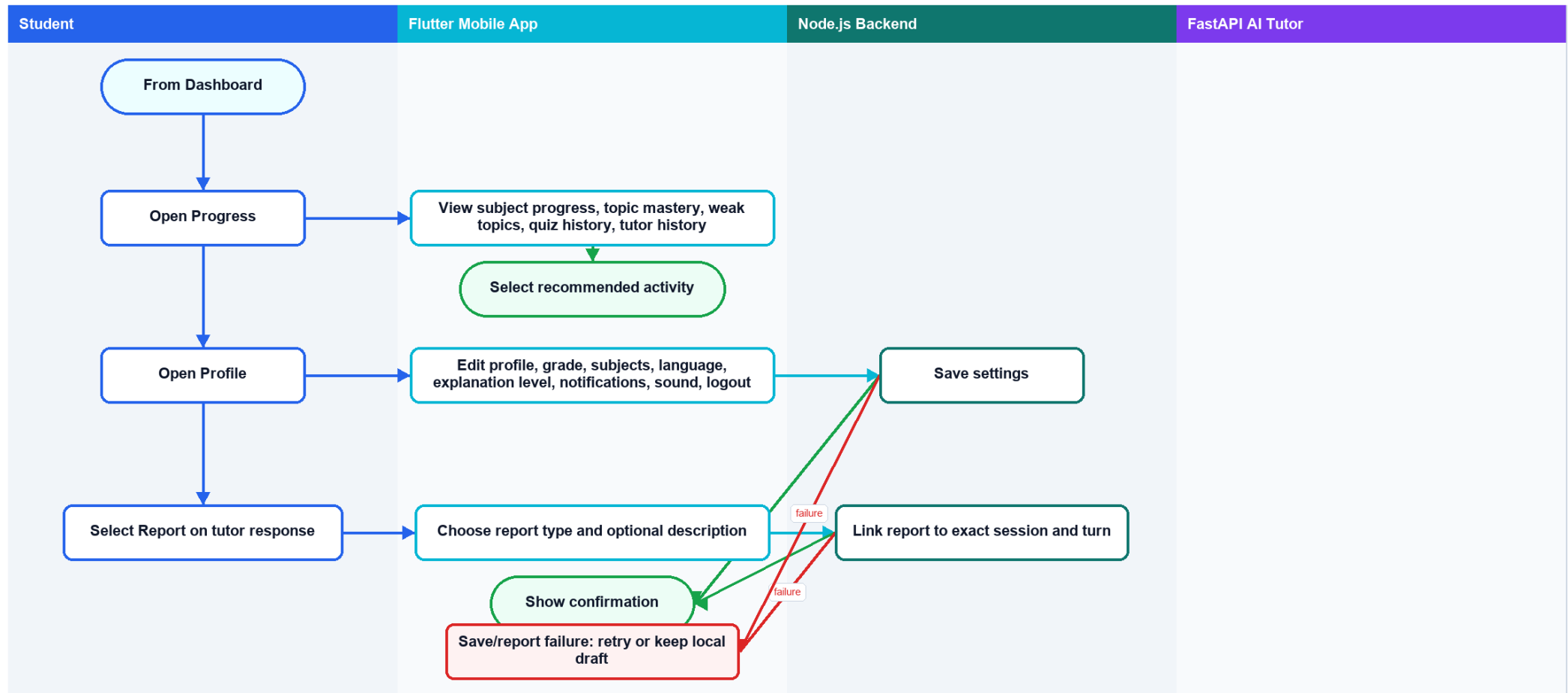


Implementation notes

- Advanced pronunciation scoring is excluded from MVP active flow.
- Raw audio should be temporary unless policy requires retention.
- Once transcript is accepted, reuse the same Instant Visual Tutor loop.

12. Profile, Progress, And Reporting

Students can review learning status, adjust preferences, and report exact tutor responses.

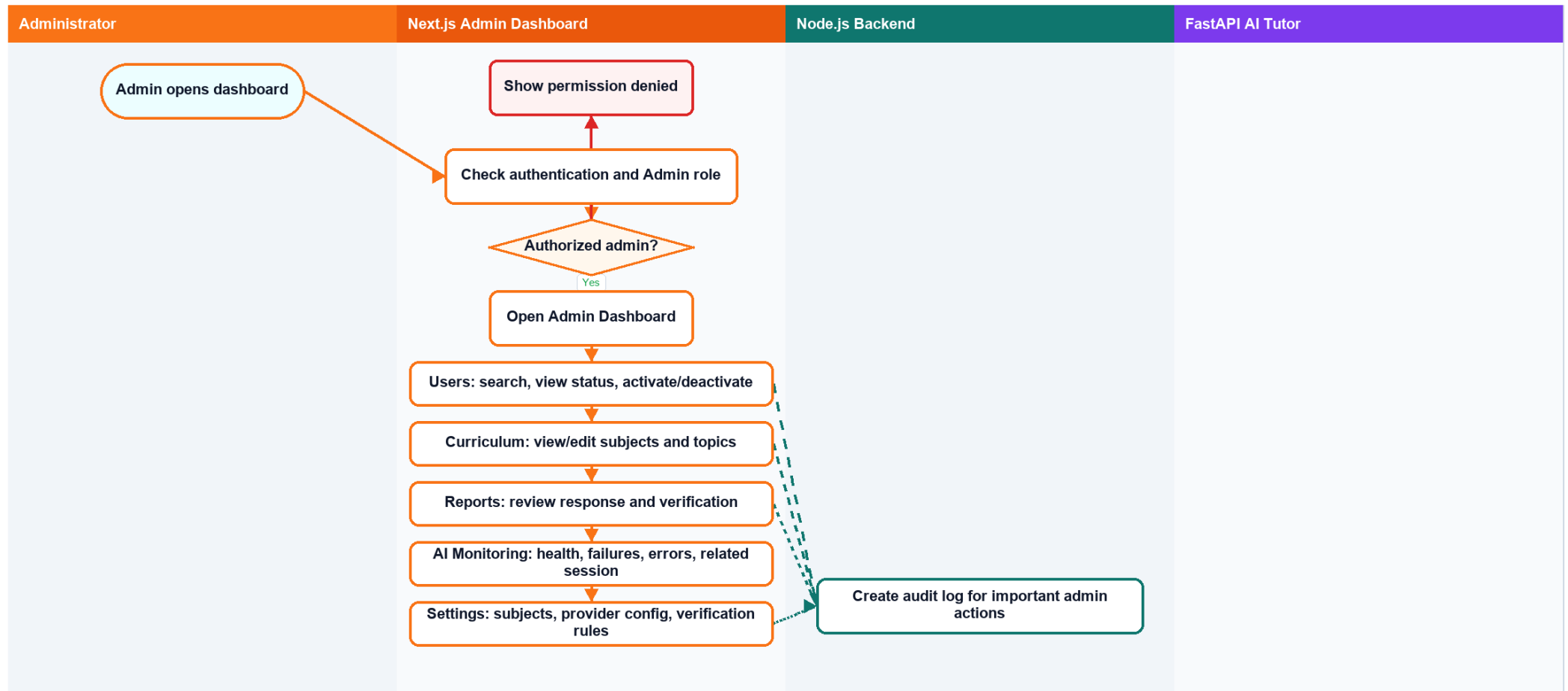


Implementation notes

- Report types include incorrect answer, unclear explanation, broken visualization, repeated response, inappropriate content, unsupported question, and voice issue.
- Reports must preserve exact tutor session and tutor turn identifiers.
- Profile settings should not interrupt active tutoring sessions.

13. Simplified Administrator Flow

Essential management and AI quality review only; every important action creates an audit log internally.

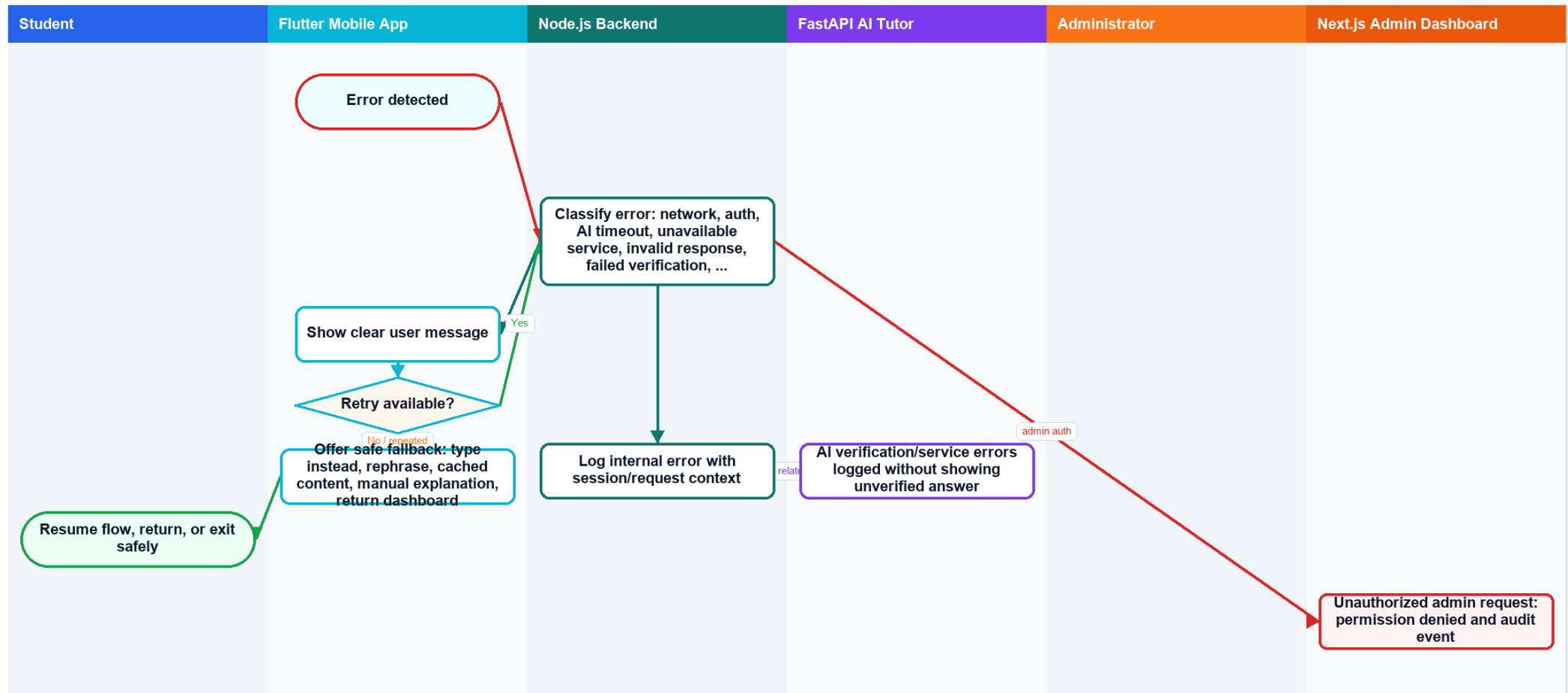


Implementation notes

- No parent, teacher, multi-level admin, escalation, or complex assignment workflow is included.
- Admin access requires the Administrator role.
- AI monitoring opens related tutor sessions for practical review.

14. Error And Recovery Flow

Every error path provides a clear message, retry, safe fallback, return action, and logging where appropriate.



Implementation notes

- Avoid dead ends: every failure returns to retry, fallback, dashboard, or safe exit.
- Invalid structured responses and failed verification must not reach students as confident answers.
- No-internet mode can preserve drafts but does not provide offline AI tutoring.

15. Legend, Assumptions, And Excluded Scope

Legend and scope boundaries for implementation and presentation.

